



ELSEVIER

Journal of
Environmental Radioactivity 59 (2002) 329–350

JOURNAL OF
ENVIRONMENTAL
RADIOACTIVITY

www.elsevier.com/locate/jenvrad

Incorporating soil structure and root distribution into plant uptake models for radionuclides: toward a more physically based transfer model

Achim Albrecht^{a,*}, Ute Schultze^b, Markus Liedgens^a,
Hannes Flühler^b, Emmanuel Frossard^a

^a*Institute of Plant Sciences ETH Zürich, Experimental Station Eschikon, Postfach 185, Eschikon 33, CH 8315 Lindau, Switzerland*

^b*Institute of Terrestrial Ecology, ETH Zürich, Grabenstrasse 11, CH 8952 Schlieren, Switzerland*

Received 3 January 2001; received in revised form 20 March 2001; accepted 26 March 2001

Abstract

Most biosphere and contamination assessment models are based on uniform soil conditions, since single coefficients are used to describe the transfer of contaminants to the plant. Indeed, physical and chemical characteristics and root distribution are highly variable in the soil profile. These parameters have to be considered in the formulation of a more realistic soil–plant transfer model for naturally structured soils.

The impact of monolith soil structure (repacked and structured) on Zn and Mn uptake by wheat was studied in a controlled tracer application (dye and radioactive) experiment. We used Brilliant Blue and Sulforhodamine B to dye flow lines and ⁶⁵Zn and ⁵⁴Mn to trace soil distribution and plant uptake of surface-applied particle-reactive contaminants. Spatial variation of the soil water content during irrigation and plant growth informs indirectly about tracer and root location in the soil profile. In the structured monolith, a till pan at a depth of 30 cm limited vertical water flow and root penetration into deeper soil layers and restricted tracers to the upper third of the monolith. In the repacked monolith, roots were observed at all depths and fingering flow allowed for the fast appearance of all tracers in the outflow. These differences between the two monoliths are reflected by significantly higher ⁵⁴Mn and ⁶⁵Zn uptake in wheat grown on the structured monolith. The higher uptake of Mn can be modelled on the basis of radionuclide and root distribution as a function of depth and using a combination of preferential flow and rooting. The considerably higher uptake of Zn requires transfer factors

*Corresponding author. Tel.: +41-52-354-9286; fax: +41-52-354-9119.

E-mail addresses: achim.albrecht@ipw.agrl.ethz.ch (A. Albrecht).

which account for variable biochemical uptake as a function of location. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords: Radioactivity; ^{65}Zn ; ^{54}Mn ; Plant uptake; Soil; Roots; Model

1. Introduction

The soil-to-plant transfer factor, the ratio between the radionuclide activity measured in the soil and the surface parts of the plant and which is used in a large number of biosphere and contamination assessment models (see Renaud, Stapel, Maubert, Bleher, & Wirth, 1999 and references therein) relies on the assumption of homogeneous distribution in the soil. The relevance of the soil-to-plant transfer process extends from radioecology (Müller & Pröhl, 1993; Renaud et al., 1999) to soil quality assessment (McBride, 1998). Radionuclides serve as model substances not only in the field of radioecology but also in stable metal research (e.g. Sinaj, Mächler, and Frossard (1999) for Zn, Echevarria, Morel, Fardeau, and Leclerc-Cessac (1998) for Ni or Gérard, Echevarria, Sterckeman, and Morel (2000) for Cd). The general modelling approach relies on the original equation

$$C_{\text{plant}} = C_{\text{soil}}F, \quad (1)$$

where the plant concentration (C_{plant}) is expressed in terms of the bulk concentration in the soil (C_{soil}) and the transfer factor F . The transfer factor can be obtained by homogeneously mixing radionuclides into a soil (Ng, 1982; Smolders, van den Brade, & Merckx, 1997) or by mixing the radionuclides into a thin surface layer (Gerzabek, Strebl, & Temmel, 1998), in both cases followed by subsequent plant growth, harvest and radioanalysis. For agricultural land, radionuclides are mixed into a soil layer of 0–20 cm (IUR, 1989), 20 cm (HSK, 1997) or 25 cm thickness (Müller & Pröhl, 1993), depending on the assumed depth of tilling. The model thus assumes that tilling will be performed after an accident and that complete homogenisation will be achieved. This is based on the concept that tilling represents an effective measure to reduce soil activity and direct exposure. This practice has proven to be counter-productive in certain areas contaminated by the Chernobyl accident, where removal of the topmost 5–10 cm served as a reliable decontamination method (Fogh et al., 1999). Enhanced wind drift and related soil erosion is furthermore an undesirable consequence, as it enhances both contaminant concentration in the air and the spread of contamination (case of As in southern Spain; CSIC, 1999). The development of a model which does not rely on tilling, but which takes the measured depth distribution into account, has thus further merit.

In a recent paper we have demonstrated the existence of a relationship between soil structure and the distribution of atmospherically deposited radionuclides and roots (Bundt, Albrecht, Froidevaux, Blaser, & Flühler, 2000). In this particular case of a forest stand, both radionuclides (^{137}Cs , ^{210}Pb , Pu and Am) and roots are enriched in the vicinity of preferential flow paths which can be mapped as a function of depth and opposed to the soil matrix (Flury, Flühler, Jury, & Leuenberger, 1994).

We have developed a model that considers the location of radionuclides (physical parameter) and roots (biological parameter) both as a function of depth and soil hydrology (soil matrix and preferential flow paths). The objective of the present publication is to test the validity of the proposed model and to evaluate the possibility of measuring or estimating the required parameters. For this purpose, wheat uptake of ^{65}Zn and ^{54}Mn was studied in soil monoliths with different structures. In a first step, we evaluated the effect of a till pan on root and radionuclide distributions and their consequences on soil–plant transfer.

2. Materials and methods

An undisturbed monolith and sufficient soil for the assembly of a repacked monolith were taken on December 1, 1999 from agricultural land used for vegetable production (Table 1). A lysimeter casing made of steel plates (4 mm thick) with an area of 30 cm × 40 cm and a depth of 70 cm was driven into the soil with a hammer. To facilitate depth penetration, the metal edge was sharpened at the lower end and the surrounding soil was removed using a small excavator. The soil block was separated from the underlying soil using a cutting plate, which was horizontally inserted. Then we tilted and lifted the soil block. The lower boundary of the monolith was leveled with a knife and protected with a wooden plate for transport. A till pan with higher density and reduced hydraulic conductivity developed at a depth of 30 cm as a consequence of intensive traffic (Bruckler, Lafolie, & Tardieu, 1991). In the disturbed soil, which was air-dried, sieved using 2 mm mesh size and repacked to original stratigraphy in a plastic-covered wooden box of the same

Table 1

Characteristics of the Obfelden soil. No continuous sampling was carried out because an earlier study at the same site (Flury et al., 1994) did not indicate significant depth variability^a

Depth (cm)	Structured density			Repacked density		For both monoliths						
	(g/cm ³) 1991 ^b	Log 1999	Log K_s^c	(g/cm ³) 1999	Log K_s^c	θ^d at sampling	Total C (%)	N (%)	Sand (%)	Silt (%)	Clay ^c (%)	Type and class
2–12	1.38	1.46	1.6 ± 0.2	1.43	1.7 ± 0.2	0.41	4.7	0.2	25	70	5	Medium crumb, 2–5 mm
22–32		1.52	1.5 ± 0.3	1.39	1.8 ± 0.2	0.37	4.8	0.21	28	68	4	Medium crumb, 2–5 mm
52–62	1.37	1.38	1.8 ± 0.2	1.42	1.7 ± 0.2	0.42	3.2	0.06	17	78	5	Coarse angular blocky, 20–50 mm

^aStructure grade is moderate at all depths.

^bFlury et al. (1994).

^cModelled implementing pedotransfer functions (PTFs) based on artificial neural networks (Schaap et al., 1998).

^dVolumetric water content (cm³/cm³).

^eBased on sedigraph analysis, after 6 min of ultrasound treatment in 0.1% Na-hexametaphosphate.

dimensions as the housing of the structured monolith, the till pan did not last. Our objective was to compare two soil monoliths essentially varying in soil structure. Both monoliths were mantled with 14 cm thick Styrofoam to shield the soil from short period temperature variation and placed onto an inclined container partly filled with quartz and equipped with an outflow system to collect the soil leachate. The repacked lysimeter was drop irrigated for one month to re-establish the original water content. Both monoliths were then stored in a greenhouse at 13°C and 15°C night and day time temperature.

2.1. Controlled irrigation

Controlled irrigation was carried out using a sprinkling device manufactured at the Institute of Terrestrial Ecology, ETH Zürich (Flury et al., 1994). The two monoliths were successively irrigated at low and high intensity. The low intensity irrigation regime (0.43 ± 0.7 cm/h during a period of 489 h) was used to apply a solution containing a fluorescent dye (Sulforhodamine B) and radiotracers (^{58}Co , ^{60}Co). It was carried out to study the flow characteristics of the two lysimeters (Albrecht, Schultze, Bello Bugallo, Wydler, Frossard, & Flühler, 2001). The high intensity irrigation (0.73 ± 0.02 cm/h) was carried out during a period of 5.4 h on March 21, 2000. The solution of deionised water, which carried a dye tracer (0.81 mmol/l Brilliant Blue, Flury & Flühler, 1994) and two radiotracers (14 kBq/l ^{54}Mn and 11 kBq/l ^{65}Zn , both in chloride form and diluted from carrier-free solutions purchased from Amersham®) had a conductivity of 0.02 mS/cm and a pH of 6.7. After a period of 2.15 h and an application of 1.9 l, ponding occurred in the structured monolith. As a consequence, irrigation was continued only on the repacked monolith, resulting in a higher volume of applied solution (4.7 l) and higher application of both dye and radiotracer.

2.2. Seeding and plant growth

On March 27, 6 days after the intensive irrigation, seedlings of wheat (*Triticum aestivum* L.) were planted in three rows perpendicular to the long side of the monolith: One in the centre, 20 cm from the edges, the other two at a distance of 6.5 cm from the edges. The target plant density was 420 plants/m² (18 plants/row). To guarantee a uniform crop, the seeding rate was twice as high and the plant stand was thinned to the planned plant density on April 5, 2000. The removed plants were collected, grouped to one sample per monolith, dried and prepared for γ -spectroscopic analysis. During growth, plants were supplied at intervals with 200 ml nutrient solution (50 mg N and P₂O₅ and 37.5 mg K₂O), with a total of 3.6 l per monolith. Wheat plants were harvested when they reached the 7-leaf stage. As wheat growth on the repacked monolith was faster, harvest was earlier (37 days after seeding) compared to the undisturbed monolith (46 days). For both monoliths, wet mass, leaf area and length for each plant were measured, followed by freeze-drying and pulverisation. To obtain the necessary dry plant volume, we grouped 3 individual plants for γ -spectroscopic analysis.

2.3. Measurement of physical parameters during irrigation and plant growth

The water content was determined using time domain reflectometry (TDR, Roth, Schulin, Flühler, & Attinger, 1990). TDRs were inserted into the repacked monolith at 4 different depths (5, 20, 33 and 56 cm from the soil surface). The metal housing of the structured monolith did not permit the installation of horizontal TDRs and hence measurement of water content. We obtained readings using a TDR installed vertically from the soil surface during the entire irrigation and growth period. Information about the water content at the bottom of the monolith was obtained from a second TDR introduced from below towards the end of the growth period. The rods have a length of 15 cm, the two vertically installed TDRs thus give average readings from 0 to 15 cm (surface TDR) and 54 to 69 cm depth (bottom TDR). Measurements were collected continuously using a cable tester and a data logger. Preferential flow developed along the upper vertical rod but was limited to its length.

2.4. Collection of monolith outflow

Both lysimeter bases had an outlet for outflow collection. Outflow was present only in the case of the homogenised monolith and was collected at the end of the irrigation (I3-H1) and during the night following irrigation (I3-H2). For both samples, the total volume was recorded and the activities of ^{54}Mn and ^{65}Zn were measured using γ -spectroscopy.

2.5. Monolith dismantling and soil sampling

After plant harvest, both monoliths were dismantled and samples collected as a function of location and depth. One group of samples represents the surface layer of 4 cm thickness, a second the edge of the monolith and the third the central part. Edge and centre samples were sub-sampled at four depth intervals: 4–19, 19–34, 34–52 and 52–65 cm. In the central portion some samples were further separated according to dye distribution patterns. The presence of coloured dyes specifies active flow patterns, their absence indicating the soil matrix (Bundt et al., 2000; Kung, 1990). Furthermore, bulk soil samples were taken from soil cores to analyse radionuclide depth migration. All samples were manually homogenised and prepared for γ -spectroscopy.

2.6. γ -Spectroscopy

All samples were analysed for ^{65}Zn (half-life 243.9 days, characteristic line at 1115.55 keV) and ^{54}Mn (half-life 312.2 days, 834.84 keV) at the γ -spectrometry laboratory of EAWAG on high purity Ge detectors. Plant samples with volumes between 1 and 10 ml were measured in bore-hole geometry, while outflow and soil samples were either measured with bore-hole or flat crystals according to sample volume (1–10 or 10–500 ml). Radionuclide activities were determined for wheat and soil samples in Bq/kg (dry weight) and for solutions in Bq/l, decay-corrected to the

common date of February 18, 2000 12:00. Geometry correction and calibration are based on standard solutions.

2.7. Root separation

We employed two methods to measure root mass as a function of depth. Soil samples ranged from 0.5 to 2 kg dry weight and represented the 4 depth segments. In one method roots were separated from the mineral fraction of the soil based on their lower specific weight. After placing the sample into 15 l buckets filled with water, the mixture was vigorously stirred; organic soil constituents including roots with a density smaller than 1 g/cm^3 migrated to the surface or remained in suspension. The suspended load was then filtered using a $40 \mu\text{m}$ mesh sieve, then solid substances were dried and roots handpicked. Using a second less labour-intensive method, we separated organic substances based on their size by rinsing the soil sample through a 0.5 mm mesh-size funnel and removing the inorganic soil fraction by density separation in a 2 l plastic beaker. The concentrate of organic material collected using the second method was not handpicked. The organic concentrate collected with both methods was freeze-dried and weighed, with concentrations given in mg per mass of dry soil.

3. Results

3.1. Water content and outflow

No outflow occurred in the structured monolith. The water content in the upper 15 cm increased from 44% to 54%, the saturation point (Fig. 1). A 10% increase in the water content of the upper 15 cm (with a volume of $18,000 \text{ cm}^3$) reflects storage of 1.8 l, thus 95% of the total irrigated volume, indicating that water storage occurred predominantly in the upper 15 cm. At this water content, ponding intensified and irrigation had to be stopped. The hydraulic situation in the repacked monolith was considerably different. The water content of the upper section increased to a maximum of 52%. Ponding occurred here only locally, because water was able to move to deeper soil layers, as can be seen from the increasing water content at all soil depths (Fig. 1). At 56 cm depth, no breakthrough occurred but a continued increase in water content was observed even after irrigation stopped. This phenomenon is related to the lower boundary condition, the transition from silty loam to coarse quartz, where the matrix potential exceeds gravitational forces. This provoked stagnation. Outflow occurred at the bottom of the repacked monolith after 225 min of irrigation, but outflow remained below irrigation intensity. Brilliant Blue and both radionuclides were found in the outflow, with Brilliant Blue reduced from $8.1 \times 10^{-4} \pm 0.03 \text{ mol/l}$ (input solution) to $5.0 \times 10^{-4} \pm 0.02$ (S1) and $6.1 \times 10^{-4} \pm 0.02 \text{ mol/l}$ (S2, 2 outflow samples). Only traces of ^{54}Mn and ^{65}Zn were found in the outflow (51 ± 1 and $31 \pm 6 \text{ Bq/l}$, respectively) as compared to the input solution ($> 11,000 \text{ Bq/l}$ for both radionuclides, Table 2). This confirms the much

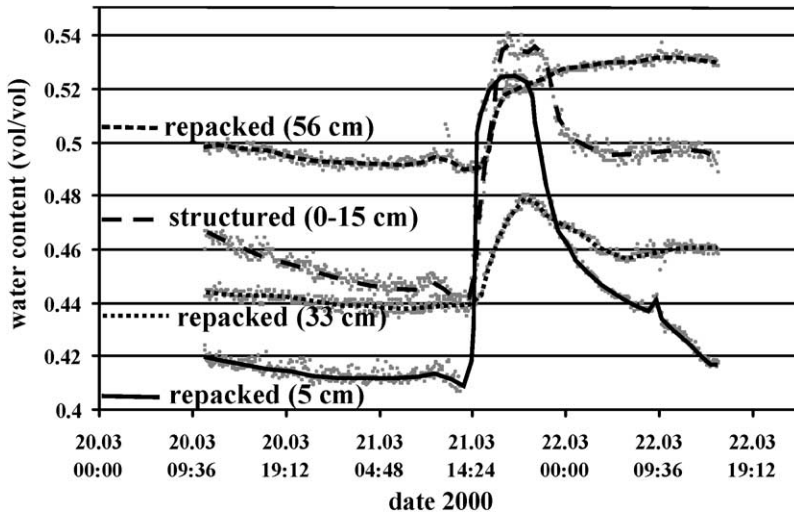


Fig. 1. Development of the water content during irrigation for the repacked monolith at depths 5 cm (H.B1), 33 cm (H.B3) and 56 cm (H.B4) and for the structured monoliths at a depth representing 0–15 cm (N.B1). The repacked monolith (H) received 4.7 l, the structured (N) 1.9 l of solution.

Table 2

Concentration of Brilliant Blue, ^{65}Zn and ^{54}Mn in the outflow of the repacked monolith

(A) Structured monolith	No outflow		
(B) Repacked monolith			
Sample ID	Input solution	I3-H1	I3-H2
Sampling start		21.3.2000 18:05	21.3.2000 19:45
Sampling end		21.3.2000 19:45	22.3.2000 7:30
Outflow volume (ml)		965 ± 48	450 ± 23
Conductivity (mS/cm)	0.02	0.31	0.41
pH	6.7	7.4	7.4
^{54}Mn (Bq/l)	11,425 ± 409	52.3 ± 5.9	49.1 ± 5.1
^{65}Zn (Bq/l)	14,390 ± 528	38.0 ± 4.8	25.0 ± 3.2
Brilliant Blue (mmol/l)	0.81 ± 0.03	0.50 ± 0.02	0.61 ± 0.02

higher mobility of Brilliant Blue relative to the two metals and the slightly higher mobility of Mn relative to Zn in soils (Flury & Flühler, 1995; Shuman & McCracken, 1999).

The water content in both monoliths dropped continuously during the wheat growth period (Fig. 2). The supply of the nutrient solution increased the water content of the surface layer of both monoliths (5 and 0–15 cm) but had no consequence on the water content at greater depth. The surface layer of the repacked monolith went through three different phases, a first phase characterised by high water losses, that lasted about 5 days; a second phase with lower water losses lasting for one month and a final phase lasting 6 days with only minor reduction in the

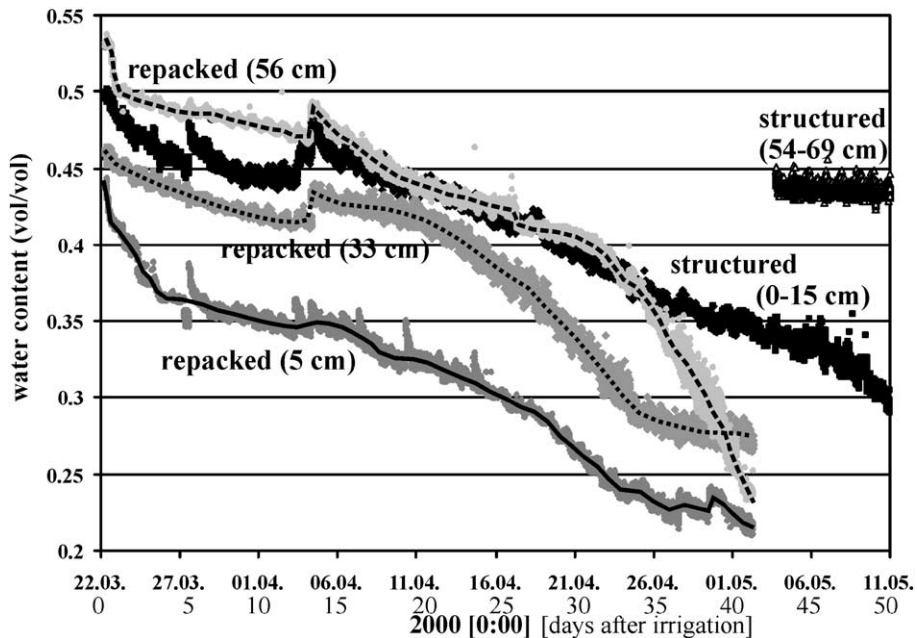


Fig. 2. Development of the water content for the repacked (at three depths) and structured monoliths (representing 0–15 and 54–69 cm depth) during plant development.

water content. A similar trend can be seen at the intermediate depths, with the only exception of a lower gradient in water loss during the first phase. The trend at the greatest depth shows a noteworthy increase in the gradient starting April 23, 2000. The water content in the top soil layer of the structured monolith is higher compared to the repacked monolith and losses occur at a lower rate. Although direct comparisons at other soil depths are not possible, it is remarkable that the water content at the bottom of the structured soil is almost twice as high during the final wheat growth period in the structured compared to the unstructured soil.

3.2. Wheat development and activity

Plant development was retarded in the structured compared to the repacked monolith. Leaf growth stage 7 was observed 7 days later and growth was less (reduced fresh weight, leaf area and length, Table 3). The mass-related activity of ^{65}Zn was much higher in wheat grown on the structured ($1910 \pm 590 \text{ Bq/kg}$) compared to repacked soil ($480 \pm 160 \text{ Bq/kg}$, Table 4). The differences in mass-related activity of ^{54}Mn between the structured ($106 \pm 38 \text{ Bq/kg}$) and repacked soils ($74 \pm 31 \text{ Bq/kg}$) were smaller.

The total uptake of ^{54}Mn and ^{65}Zn per plant normalised to the applied volume (Fig. 3) accounts for the difference in the volume of tracer solution and the difference in plant mass. The different uptake of wheat on structured soil is striking, the

Table 3

Information relating to plant development at the time of harvest (3.5.2000 for repacked monolith; 12.5.2000 for structured monolith). Most plants reached the 7-leaf stage, in 35% of wheat grown on the structured monolith ear appearance had initiated. The ranges indicate standard deviations

	Repacked monolith			Structured monolith		
	Fresh weight (g)	Total area (cm ²)	Total length (cm)	Fresh weight (g)	Total area (cm ²)	Total length (cm)
First row	2.2±0.4	73.7±9.3	137.9±14.9	1.2±0.4	25.4±8.4	74.9±22.7
Second row	1.9±0.5	71.4±13.2	111.6±15.2	1.2±0.4	25.6±6.3	72.1±10.6
Third row	2.0±0.5	71.6±15.1	109.0±14.2	1.2±0.5	24.4±8.3	66.8±12.0
Overall	2.0±0.5	72.2±12.9	119.5±19.9	1.2±0.4	25.2±7.7	71.5±16.7

Table 4

Average activity and normalised uptake of wheat plants (3 individual plants grouped in one sample) collected from 3 rows on the structured and repacked monoliths. Normalisation to the irrigation volume is necessary because it was higher on the repacked (4.7l) than on the structured monolith (1.9l). 2σ values represent individual measurement errors. The range given on average values indicate standard deviations based on the entire population as an argument $\left(\sqrt{(n \sum x^2 - (\sum x)^2)/n^2}\right)$

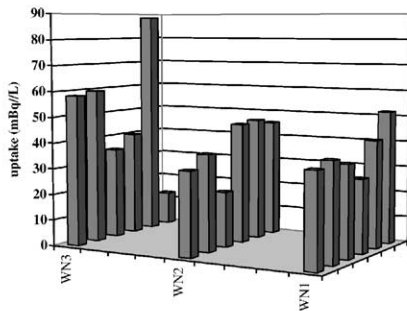
		Plant mass (g)	⁵⁴ Mn				⁶⁵ Zn			
			Activity (Bq/kg)	2σ	Uptake (mBq/l)	2σ	Activity (Bq/kg)	2σ	Uptake (mBq/l)	2σ
<i>(A) Repacked monolith</i>										
WH1_1–3	1.3	66	14	18	2	346	59	39	7	
WH1_3–6	1.2	54	13	13	1	342	62	34	6	
WH1_7–9	0.9	38	14	8	1	242	54	21	5	
WH1_10–12	1.2	53	11	13	1	354	56	37	6	
WH1_13–15	1.4	85	13	25	2	392	58	48	8	
WH1_16–18	1.2	141	21	37	2	847	114	92	13	
WH2_1–3	1.4	64	13	18	2	390	63	47	8	
WH2_4–6	1.1	31	10	7	1	264	50	26	5	
WH2_7–9	1.1	54	13	12	1	604	93	57	9	
WH2_10–12	0.7	43	14	7	1	494	86	32	6	
WH2_13–15	0.8	64	14	10	1	412	71	27	5	
WH2_16–18	1.2	85	16	21	2	642	94	65	10	
WH3_1–3	1.3	82	15	21	2	509	77	55	9	
WH3_4–6	1.0	86	16	18	1	453	74	40	7	
WH3_7–9	1.0	147	24	31	2	608	93	52	8	
WH3_10–12	1.1	58	12	15	1	431	67	45	7	
WH3_13–15	1.2	83	14	21	1	505	73	53	8	
WH3_16–18	1.0	99	18	20	2	765	109	63	10	
Average		74.0±30.6		18±8		478±159		46±17		
<i>(B) Structured monolith</i>										
WN1_1–3	0.9	73	17	36	8	1225	169	600	87	
WN1_4–6	0.6	114	24	38	8	1826	255	608	88	
WN1_7–9	0.7	102	21	35	8	1713	233	595	85	
WN1_10–12	0.6	81	20	28	7	1942	262	679	95	
WN1_13–15	0.9	89	17	42	8	1710	220	800	109	

Table 4 (continued)

	Plant mass (g)	⁵⁴ Mn				⁶⁵ Zn			
		Activity (Bq/kg)	2σ	Uptake (mBq/l)		Activity (Bq/kg)	2σ	Uptake (mBq/l)	2σ
WN1_16–18	1.0	100	18	52	9	2031	260	1053	142
WN2_1–3	0.8	75	15	32	7	1386	182	601	83
WN2_4–6	1.0	71	15	37	8	1352	176	711	99
WN2_7–9	1.1	38	11	21	6	1163	152	649	91
WN2_10–12	0.9	96	17	47	9	1684	218	816	113
WN2_13–15	0.6	141	25	47	8	2909	363	979	128
WN2_16–18	0.1	119	24	46	9	2631	335	1003	135
WN3_1–3	1.1	100	18	58	11	1577	206	918	128
WN3_4–6	0.9	124	20	59	10	1918	244	920	123
WN3_7–9	0.4	176	38	35	7	3091	414	618	83
WN3_10–12	0.8	92	18	40	8	1632	211	713	98
WN3_13–15	0.8	199	29	88	14	2634	333	1170	156
WN3_16	0.3	76	29	13	5	1259	208	213	35
WN3_17–18	0.3	153	50	20	6	2569	404	344	50
Average		106 ± 38		41 ± 16		1908 ± 574	573.5	736 ± 232	

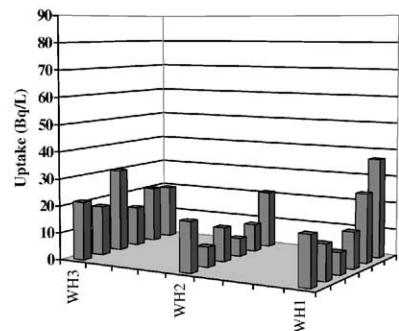
Structured monolith

⁵⁴Mn average uptake = 41 ± 16 mBq/L

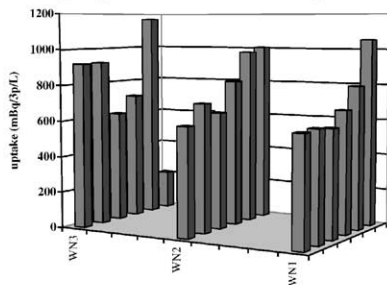


Repacked monolith

⁵⁴Mn average uptake = 18 ± 8 mBq/L



⁶⁵Zn average uptake = 736 ± 232 mBq/L



⁶⁵Zn average uptake = 46 ± 17 mBq/L

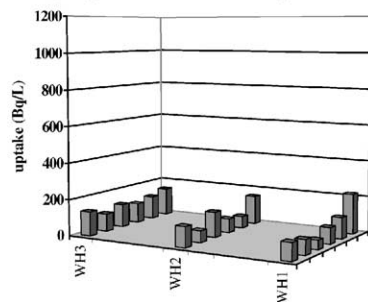


Table 5

Average activities of ^{65}Zn and ^{54}Mn in the structured (A) and repacked (B) monolith as a function of depth and location. Variability indicates the standard deviation or the range of measurements

Compartment	Sampling depth (cm)		⁵⁴ Mn (Bq/kg)	Variability	⁶⁵ Zn (Bq/kg)	Variability
	From	To				
(A)						
N_surface1	0	4	1837	154	1563	144
N_surface2	4	6	410	49	329	47
N_edge1	6	19	67.3	15.3	48.2	11.5
N_edge2	19	34	59.4	5.2	42.6	4.1
N_edge3	34	52	2.8	0.5	3.1	0.7
N_edge4	52	65	0.5	0.2	<1.4	
N_no_dye1	6	16	5.1	1.9	17.5	12.8
N_no_dye2	16	28	<4.7		14.2	5.6
N_no_dye3	28	44	<6.8		<14.1	
N_no_dye4	44	64	<1.5		<1.4	
N_dye1	6	16	63.2	17.3–194	59.3	22.9–137
N_dye2	16	28	70.5	8.3–211	65.0	18.2–230
N_dye3	28	44	49.4	<2.2–96.5	51.5	<6.1–65.2
N_dye4	44	64	<1.5		<1.4	
N_bottom	65	69	<1.7		<4.5	
(B)						
H_surface	0	4	1905	211.1	1546	201
H_edge1	4	19	326.5	26.4	278	24.3
H_edge2	19	34	45.9	4.1	47.0	4.8
H_edge3	34	52	37.2	4.6	41.3	4.7
H_edge4	52	65	1.6	0.5	4.8	0.0
H_dye1	No separation dye—no-dye					
H_dye2	17	35	220.1	4.5–418	226	10.7–360
H_dye3	35	50	366.6	121–630	373	128–620
H_dye4	53	65	4.8	<2.8–6.7	<9.2	2.9
H_no_dye1	No separation dye—no-dye					
H_no_dye2	17	35	<2.1		<5.4	
H_no_dye3	35	50	<5.3		<5.4	
H_no_dye4	50	65	<2.5		<7.2	
HD Boden	65	69	<1.5		<3.9	

uptakes of ^{54}Mn and of ^{65}Zn being higher by factors of 2.3 and 15.9 in the structured monolith. The variation in plant uptake within and between rows is occasionally large (e.g. for the repacked monolith: 7 vs. 37 mBq/l for ^{54}Mn and 63 vs. 220 mBq/l for ^{65}Zn) but it is smaller than the difference between the two soil types.



Fig. 3. Uptake of ^{65}Zn and ^{54}Mn by wheat planted on structured (left) and repacked (right) monoliths. Columns indicate the total uptake of 3 plants normalised to 1 l of input solution. The distribution of bars mirrors the position of wheat plants on the surface of the monolith.

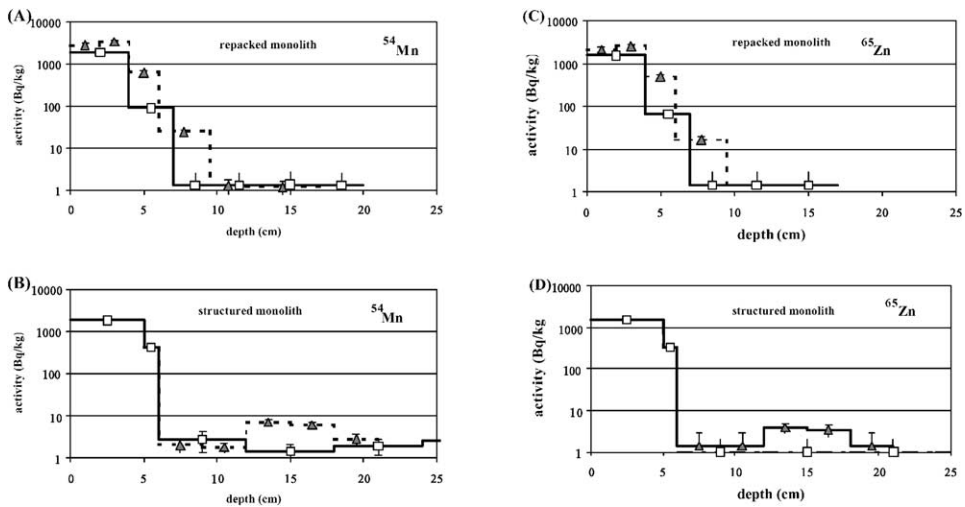


Fig. 4. Distributions of ^{54}Mn (A, B) and ^{65}Zn (C, D) in two profiles sampled in the repacked monolith (upper diagrams) and the structured monolith (lower diagrams). Soil cores were taken at least 10 cm away from the monolith edge. Error bars indicate the 2σ measurement error or the detection limit.

3.3. Soil activity

The mass-related activities of ^{65}Zn and ^{54}Mn in soil samples from both monoliths (Table 5) show a number of characteristic differences: (1) samples collected at the edge of the monolith have higher activities of both ^{65}Zn and ^{54}Mn compared to average activities in the centre of the monoliths; (2) dyed soil samples collected in the centre of the monolith have a higher activity than samples without dye and (3) measurable activities of both radionuclides can be found at greater depth in the repacked monolith than in the structured monolith.

The detailed depth distribution of both radionuclides in the upper 25 cm (Fig. 4) indicates mostly differences related to monolith structure. In the repacked monolith, the activities of ^{54}Mn and ^{65}Zn decrease continuously, without significant differences between the two metals. In the structured monolith, the depth distribution is more irregular, with a zone, sampled in one profile, indicating local ^{54}Mn and ^{65}Zn enrichment at 15 cm depth. In bulk soil samples collected at greater depth (no dye; Table 5), radionuclides remained near or below the detection limit, higher activities being recorded in samples collected near the edge and within dyed flow zones of both monoliths (edge, dye; Table 5).

3.4. Distribution of roots and/or solid organic matter

The relative distribution of both roots and solid organic matter, as a function of depth (Fig. 5) indicates a higher depth penetration of roots in the repacked monolith. The proportion of roots found at a depth greater than 34 cm is lower in the

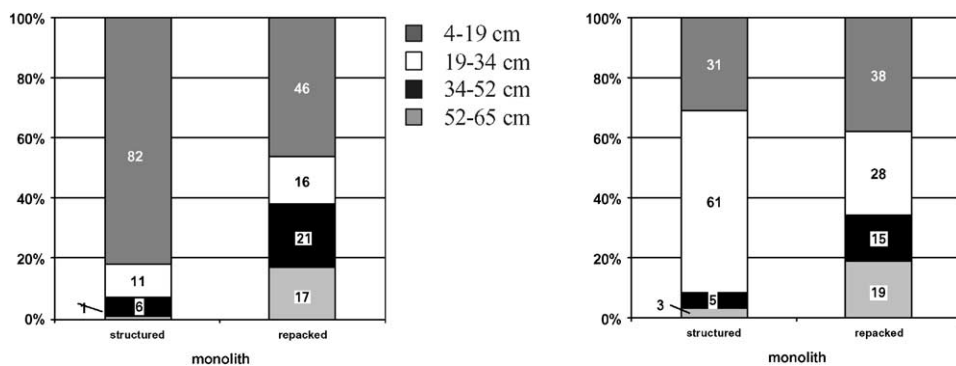


Fig. 5. Relative mass distributions of roots and/or organic matter in soil samples from 4–19, 19–34, 34–52 and 52–65 cm depth taken in the structured and repacked monoliths. Results shown in the diagram on the right were obtained by density separation followed by hand picking, those on the left by combined size and density separation without hand picking.

structured (7–8%) compared to the repacked monolith (38% or 34%). Compared to the undisturbed monolith, the repacked monolith had higher absolute root dry weight density in the top soil (39.8 vs. 23.5 mg/kg_{dry soil}) and in deeper soil layers (28.0 vs. 2.9 mg/kg_{dry soil}).

4. Discussion

4.1. The impact of soil physics on tracer and root distribution

The till pan present in the structured monolith at a depth between 30 and 35 cm had a most significant impact on soil hydraulics, solute transport and root distribution. It limited both infiltration and evaporation in the structured monolith. As a consequence, the structured monolith had no outflow (Table 2) and the water content in the top soil reached saturation within less than 2 h (Fig. 1). The entire volume of added solution (including all tracers) was stored in the upper 30 cm of the soil profile. The number of roots that were able to penetrate the dense layer is small and the resulting root density below the depth of 30 cm remained insignificant. The relatively high water content at depth in the structured monolith (Fig. 2) indicates reduced capillary rise during evaporation, again caused by the low hydraulic conductivity of the till pan. The repacked monolith is characterised by higher hydraulic conductivity and thus higher infiltration. Transport rates increased accordingly. At the bottom of the monolith (lower boundary), the transition of silty loam to sandy quartz caused water accumulation (Fig. 1, H_B4).

The fast passage of the tracer solution through the repacked monolith and the appearance of colour and radionuclide tracers indicate a type of bypass flow, which is generally termed fingering flow. It is detected at high infiltration rate under both unsaturated and saturated conditions and is caused by either soil layering or

differences in the initial water content, or a combination thereof (Gish & Shirmohammadi, 1991; Oswald, Kinzelbach, Greiner, & Brix, 1997). The presence of ^{54}Mn and ^{65}Zn in the outflow shows that radionuclides, in spite of their particle reactivity, passed the 70 cm soil filter. At depth these radionuclides were detected only in zones marked by colour dyes (dye, edge; Table 5). In bulk samples, their activity remained below the detection limit, which explains the low activity in soil cores below 20 cm.

The absence of the growth-limiting pan in the repacked soil allowed roots to develop within the entire monolith volume. This information can be drawn directly from root measurement (Fig. 5) and indirectly from the development of the water content at different depths (Fig. 2). The increased water loss gradient in the deepest part of the repacked monolith which began on April 25, 2000 is likely caused by root water extraction. Replenishment of water from below the lowest TDR was not possible.

4.2. Uptake of ^{54}Mn and ^{65}Zn

In both monoliths, the distribution of radionuclides is not homogeneous. Activities are generally higher near the soil surface and within preferential flow zones along the edge and in the centre of the monolith (Table 5; Albrecht et al., 2001). In the field, such preferential flow zones existed until 1991 down to a depth of at least 1 m (Flury et al., 1994) but were sealed as a result of recent soil compaction. The entire applied solution thus remained heterogeneously distributed within the upper 30 cm of the monolith. Root distribution as a function of depth is also strongly influenced by soil structure. In the structured monolith, almost all roots are located in the upper zone between 0 and 30 cm depth, whereas roots exploited the entire soil block of the repacked monolith (Fig. 5).

The following extended equation illustrates how the concept of inhomogeneous distribution of radionuclides and preferential rooting can be incorporated into soil-to-plant transfer models:

$$C_{\text{plant}} = \sum_{i=1}^n \sum_{j=1}^2 C_j(\Delta z_i) R_j(\Delta z_i) F_j \quad (2)$$

(with $J = 1$ for compartment 1 and $J = 2$ for compartment 2) where n refers to the number of soil horizons (dimensionless), Δz to the thickness of each individual horizon (cm), C to the radionuclide concentration (Bq/kg), F to the transfer factor (dimensionless) and R to the relative root density (dimensionless) expressed as

$$R_j(\Delta z_i) = \frac{r_j(\Delta z_i) A_j(\Delta z_i)}{\sum_{i=1}^n \sum_{j=1}^2 r_j(\Delta z_i) A_j} \quad (3)$$

with r_j and A_j being the root density (g/cm^3) and the area of compartment j (cm^2), respectively. Compartment 1 may represent the preferential flow paths and compartment 2 the bypassed soil matrix. These compartments are distinguished on the basis of dye distribution in two-dimensional soil profiles. The root density in

the two compartments is weighted with the area of the respective compartment to account for the “actual” root distribution in the soil profile (Bundt et al., 2000).

To test whether root and radionuclide distribution alone explain the different uptakes of ^{54}Mn and ^{65}Zn , we applied information on root and radionuclide distributions to our extended transfer model. As detailed root and radionuclide data are not yet available for different soil compartments, we performed the calculation only with depth as changing parameter (1-compartment case; model 1). Eqs. (2) and (3) simplify accordingly:

$$C_{\text{plant}} = \sum_{i=1}^n C_i R_i F \quad \text{with} \quad R_i = \frac{r_i A_i}{\sum_{i=1}^n r_i A_i} \quad (4)$$

The root density has been calculated from those root measurements which included handpicking (Fig. 5). The average activity at each depth is based on soil measurements (Table 5) and an estimated distribution of individual samples. The chosen transfer factors, 0.9 for ^{65}Zn and 0.2 for ^{54}Mn , were taken from the literature (Müller & Pröhl, 1993). All parameters and the results for both radionuclides on both monoliths are summarised in Table 6. The modelled plant activities for wheat grown on the repacked monolith, 105 and 369 Bq/kg for ^{65}Zn and ^{54}Mn , are comparable to the measured values (74 ± 31 and 478 ± 164 , Table 4). For the structured monolith, modelled (150 Bq/kg) and measured (110 ± 40) activities for ^{54}Mn are comparable but the modelled activity of 570 Bq/kg for ^{65}Zn remains significantly below the measured activity of 1910 ± 590 Bq/kg (Fig. 6, model 1).

Our simplified model (Eq. (4)) is comparable to the model used by Wadey, Shaw, Bell, and Minski (1994), who, to our knowledge, introduced the concept of root density into soil-to-plant uptake models for radionuclides. They used mean soil specific activities in discrete soil layers, weighted according to the fractional abundance of roots. Their model proved successful for homogenised soils with radionuclide input from an artificial groundwater level at 40 and 70 cm depth. It cannot be applied for structured soils, where radionuclide and root concentrations vary within individual soil layers.

We have found significant differences in radionuclide activity in zones belonging to the same soil horizon. In both monoliths, ^{54}Mn and ^{65}Zn were enriched in dyed preferential flow zones compared to the soil matrix. We have no data yet on the difference in root density between these zones. There is literature evidence on enhanced rooting along macropores, an important soil structure that triggers preferential flow. Whiteley and Dexter (1983) studied the ability of seminal roots of pea, rape and safflower growing along cracks to penetrate the soil matrix. Particularly when the angle of the crack is within 45° of the ideal geotropic growth direction and when the soil strength is high, the roots preferentially grow along the crack. Preference of wheat and maize roots to grow along macropores was also found in pot and field experiments (Hatano, Iwanaga, Okajima, & Sakuma, 1988; Pierret, Moran, & Pankhurst, 1999; Stewart, 1997; Tardieu, 1991). Stewart (1997) postulated for wheat plants that 80% of roots were located in macropores and within a distance of 1 mm around them. Furthermore, roots shape some features of

Table 6
List of parameters entering the uptake model and results

Depth segment (cm)		Surface area A (cm ²)	Root concentration (mg/kg soil)	Root density r (g/cm ³)	Relative root density R	⁵⁴ Mn		⁶⁵ Zn	
From	To					Soil activity C_i (Bq/kg)	Plant activity C_{plant} (Bq/kg)	Soil activity C_i (Bq/kg)	Plant activity C_{plant} (Bq/kg)
<i>Repacked monolith</i>									
0	4	120	29.6	4.2E – 05	0.15	3076.0	92.5	2325.0	314
4	34	900	10.2	1.5E – 05	0.39	161.1	12.6	146.6	51
34	52	540	11.3	1.6E – 05	0.26	5.0	0.1	7.0	2
52	65	390	12.2	1.7E – 05	0.20	2.6	0.1	7.0	1
0	65	1950					105		369
<i>Structured monolith</i>									
0	6	180	24.6	3.5E – 05	0.54	1361.4	145.8	1151.6	555
6	34	840	3.4	4.9E – 06	0.35	48.8	3.4	41.6	13
34	52	540	1.6	2.3E – 06	0.10	15.4	0.3	17.9	2
52	65	390	0.2	3.3E – 07	0.01	1.0	0.0	1.4	0
0	65	1950					150		570

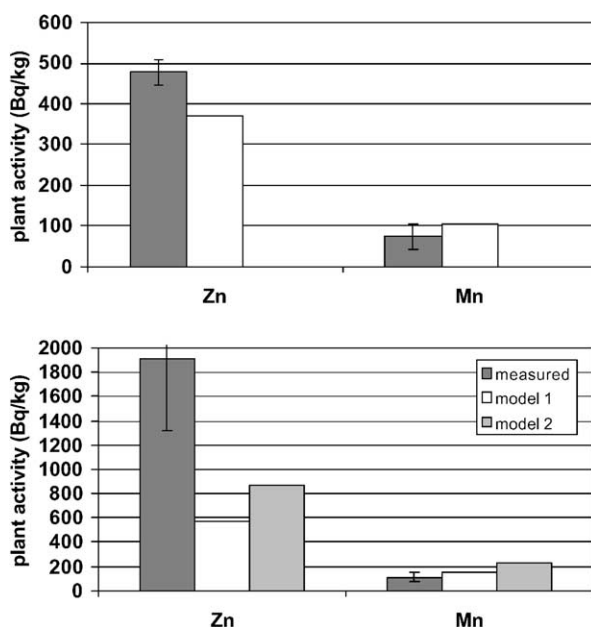


Fig. 6. Comparison of measured (with standard deviation over all measurements) and modelled plant uptake of ^{65}Zn and ^{54}Mn . For the homogenised, repacked monolith (upper diagram) only the 1-compartment model (homogenous distribution in each depth section; model 1) was used; for the structured monolith (lower diagram) a 2-compartment model was tested (model 2) which considers for each depth both heterogeneous root and radionuclide distribution.

preferential flow systems. Rotted root channels of bush beans and sun flowers, for example, act as preferential flow paths after a decay period of 4 weeks (Liu, Waldron, & Wong, 1994).

The combined effect of preferential flow and rooting in the structured soil (Fig. 6, model 2) can be evaluated using the measured concentration increase of ^{65}Zn and ^{54}Mn in flow zones compared to the soil matrix (a factor of 10) and assuming 80% of wheat roots to be located near macropores (Stewart, 1997). The relative distribution of flow paths and matrix was set equal to the results obtained by Bundt et al. (2000).

Table 7 summarises both the individual parameters used for calculation as well as the final results. The combination of preferential radionuclide distribution as a result of bypass flow and preferred rooting in flow paths causes uptake to increase by a factor of 1.5. This brings the modelled ^{65}Zn plant activity to 864 Bq/kg and the modelled ^{54}Mn uptake to 226.8 Bq/kg (Fig. 6).

In their analysis of ^{137}Cs transfer factors, Ehlken and Kirchner (1996) recognised the importance of the depth distribution of grass roots and developed a code that combines a diffusion model with a root distribution model assuming exponential root decline with depth. With the incorporation of root architecture, they were able to reproduce the change in transfer factors as a function of time to a much better degree than with a model based on meteorological results (soil moisture). Our model

Table 7

List of parameters used to model the uptake of ^{54}Mn and ^{65}Zn in structured soil with preferential flow and rooting. The value of 80% for preferential rooting of wheat was measured by Stewart (1997), enhancement of radionuclide activity in flow paths compared to matrix (factor 10) is derived from measured results summarised in Table 5 and the relative volumetric distribution of preferential flow paths (0–6 cm = 69%, 6–34 cm = 38% and 34–65 cm = 13%) was adapted from Bundt et al. (2000)

Depth segment (cm)		m = soil matrix; p = preferential flow path.	Surface area A (cm ²)	Root concentration (mg/kg soil)	Root density r (g/cm ³)	Relative root density R	^{54}Mn		^{65}Zn	
From	To						Soil activity (Bq/kg)	Plant activity (Bq/kg)	Soil activity (Bq/kg)	Plant activity (Bq/kg)
0	6	m	56	4.9	7.0E – 06	6.5E – 02	188.8	2.5	159.7	9.4
0	6	p	124	19.7	2.8E – 05	5.8E – 01	1888.3	219.2	1597.2	834.4
6	34	m	521	0.7	9.8E – 07	8.5E – 02	11.0	0.2	9.4	0.7
6	34	p	319	2.8	3.9E – 06	2.1E – 01	110.5	4.6	94.2	17.7
34	52	m	470	0.3	4.6E – 07	3.6E – 02	7.1	0.1	8.3	0.3
34	52	p	70	1.3	1.8E – 06	2.1E – 02	70.8	0.3	82.6	1.6
52	65	m	339	0.0	6.6E – 08	3.7E – 03	0.5	0.0	0.7	0.0
52	65	p	51	0.2	2.6E – 07	2.2E – 03	4.7	0.0	6.6	0.0
0	65							226.8		864.1

does not take into account the radionuclide variation or root distribution as a function of time. It considers the differences in radionuclide and root concentrations as a function of depth and, in addition to the model of Ehlken and Kirchner (1996), as a function of depth compartments.

We are aware of the complexity of biochemical processes that play a role in the soil–plant transfer. Sorption/desorption reactions (e.g. Frossard, Fardeau, Brossard, and Morel (1994) for nutrients and Sinaj et al. (1999) and Smolders, Brans, Földi, and Merckx (1999) for contaminants) or processes occurring in the rhizosphere (e.g. root exudation or mycelage, Gaume, Weidler, & Frossard, 2000; Hinsinger, 1998) can mobilise or immobilise contaminants. Furthermore, mushrooms, in particular vesicular arbuscular mycorrhiza and their hyphae systems, can enhance and reduce physical and chemical availability of nutrients and contaminants (Bolan, 1991; Jayachandran, Schwab, & Hetrick, 1992; Sharma, Srivastava, & Johri, 1994). The differences between modelled and measured results for the structured monolith (Fig. 6) clearly show site dependence of the soil–plant transfer factor. Our model provides for the possibility of changing transfer factors as a function of uptake location (matrix vs. flow paths) but no data are yet available to quantify these differences. It is nonetheless encouraging to see that a rather simple extension of the original transfer formula can make a significant contribution to modelling the soil–plant transfer in naturally structured soils.

4.3. Possibility for improvement

During our first experiment, we were not able to carry out root and radionuclide tracing to such an extent that both could be fully associated with the different flow compartments within the soil. This is related to the choice of Brilliant Blue as a flow tracer during the irrigation experiment. Brilliant Blue is relatively mobile (Flury & Flühler, 1995; Germán-Heins & Flury, 2000). A significant portion preferentially passed the monolith (Table 2) and an additional portion was remobilised during evaporation and capillary rise, the latter causing crystallisation of Brilliant Blue at the soil surface and formation of an intensively blue layer. The portion which remained sorbed within the monolith was insufficient to trace the flow paths quantitatively. A further technical difficulty relates to the existence of preferred flow and solute transport along the side of the monolith (Albrecht et al., 2001; Corwin, 2000) and the artificial lower boundary of the lysimeters (Flury, Yates, & Jury 1999; Phillips, Quisenberry, & Zeleznik, 1995). A new series of experiments, designed to eliminate lysimeter artefacts, is planned in the field. In addition, more reactive dye tracers will help to reduce tracer loss via leaching and capillary transport.

Acknowledgements

We are most grateful to Jörg Leuenberger and Hannes Wydler for technical assistance in the field and in the lab, to Pastora Bello Bugallo for helping with

modelling aspects and Dominik Bänninger for his assistance during tracer detection. The Swiss Federal Safety Inspectorate (HSK) supports AA.

References

- Albrecht, A., Schultze, U., Bello Bugallo, P., Wydler, H., Frossard, E., & Flühler, H. (2001). Distribution of surface applied dye and radio tracers in structured and repacked monoliths: Quantification of the impact of preferential flow and edge effects. *Soil Science Society of America Journal*, submitted for publication.
- Bolan, N. S. (1991). A critical review on the role of mycorrhizal fungi in the uptake of phosphorus by plants. *Plant and Soil*, 134, 189–207.
- Bruckler, L., Lafolie, F., & Tardieu, F. (1991). Modeling root water potential and soil-water transport. II. Field comparisons. *Soil Science Society of America Journal*, 55(5), 1213–1220.
- Bundt, M., Albrecht, A., Froidevaux, P., Blaser, P., & Flühler, H. (2000). Impact of preferential flow on radionuclide distribution in soil. *Environmental Science and Technology*, 34(18), 3895–3899.
- Corwin, D. L. (2000). Evaluation of a simple lysimeter-design modification to minimize sidewall flow. *Journal of Contaminant Hydrology*, 42(1), 35–49.
- CSIC (1999). Coto de Doñana; Accidente de Aznalcóllar, Vol. 1999. Grupo de expertos del consejo superior de investigaciones científicas y organismos colaboradores sobre la emergencia ecológica del Rio Guadamar, CSIC Administración Central: c/ Serrano, 117, 28006 MADRID, fax: (91)-4113077/(91)-5855001, <http://www.csic.es/hispano/coto/infor12/infor12.htm> Figura 1.1.
- Echevarria, G., Morel, J. L., Fardeau, J. C., & Leclerc-Cessac, E. (1998). Assessment of phytoavailability of nickel in soils. *Journal of Environmental Quality*, 27, 1064–1070.
- Ehlken, S., & Kirchner, G. (1996). Seasonal variations in soil-to-grass transfer of fallout strontium and cesium and of potassium in North German soils. *Journal of Environmental Radioactivity*, 33(2), 147–181.
- Flury, M., & Flühler, H. (1994). Brilliant Blue FCF as a dye tracer for solute transport studies—a toxicological overview. *Journal of Environmental Quality*, 23, 1108–1112.
- Flury, M., & Flühler, H. (1995). Tracer characteristics of Brilliant Blue FCF. *Soil Science Society of America Journal*, 59(1), 22–27.
- Flury, M., Flühler, H., Jury, W. A., & Leuenberger, J. (1994). Susceptibility of soils to preferential flow of water: A field study. *Water Resources Research*, 30(7), 1945–1954.
- Flury, M., Yates, M. V., & Jury, W. A. (1999). Numerical analysis of the effect of the lower boundary condition on solute transport in lysimeters. *Soil Science Society of America Journal*, 63(6), 1493–1499.
- Fogh, C. L., Anderson, K. G., Barkovsky, A. N., Mishine, A. S., Ponamarjov, A. V., Ramzaev, V. P., & Roed, J. (1999). Decontamination in a Russian settlement. *Health Physics*, 76(4), 421–430.
- Frossard, E., Fardeau, J. C., Brossard, M., & Morel, J. L. (1994). Soil isotopically exchangeable phosphorus: a comparison between E and L values. *Soil Science Society of America*, 58, 846–851.
- Gaume, A., Weidler, P. G., & Frossard, E. (2000). Effect of maize root mucilage on phosphate adsorption and exchangeability on a synthetic ferrihydrite. *Biology and Fertility of Soils*, 31, 525–532.
- Gérard, E., Echevarria, G., Sterckeman, T., & Morel, J. L. (2000). Cadmium availability to three plant species varying in cadmium accumulation pattern. *Journal of Environmental Quality*, 29, 1117–1123.
- Germán-Heins, J., & Flury, M. (2000). Sorption of Brilliant Blue FCF in soils as affected by pH and ionic strength. *Geoderma*, 97(1–2), 87–101.
- Gerzabek, M. H., Strebl, F., & Temmel, B. (1998). Plant uptake of radionuclides in lysimeter experiments. *Environmental Pollution*, 99, 93–103.
- Gish, T., & Shirmohammadi, A. (1991). *Preferential flow* (p. 408). St. Joseph, MI: American Society of Agricultural Engineers.
- Hatano, R., Iwanaga, K., Okajima, H., & Sakuma, T. (1988). Relationship between the distribution of soil macropores and root elongation. *Soil Science and Plant Nutrition*, 34(4), 535–546.

- Hinsinger, P. (1998). How do plant roots acquire mineral nutrients? Chemical processes involved in the rhizosphere. *Advances in Agronomy*, 64, 225–265.
- HSK (1997). Berechnung der Strahlenexposition in der Umgebung aufgrund von Emissionen radioaktiver Stoffe aus Kernanlagen, Richtlinien für schweizer Kernanlagen, HSK-R-41/d (p. 10). Hauptabteilung für die Sicherheit der Kernanlagen, Villigen-HSK.
- Kung, K.-J. S. (1990). Preferential flow in a sandy vadose soil: 1. Field observations. *Geoderma*, 46, 51–58.
- IUR. (1989). Sixth annual report of the working group on soil-to-plant transfer. International Union of Radioecologists, Grimselpass, Switzerland.
- Jayachandran, K., Schwab, A. P., & Hetrick, B. A. D. (1992). Mineralization of organic phosphorus by vesicular-arbuscular mycorrhizal fungi. *Soil Biology and Biochemistry*, 24(9), 897–903.
- Liu, I.-W. Y., Waldron, L. J., & Wong, S. T. S. (1994). Application of nuclear magnetic resonance imaging to study preferential water flow through root channels. In S. H. Anderson, & J. W. Hopmans (Eds.), *Tomography of soil-water-root processes*, Vol. SSSA Special Publication 36 (pp. 135–148). Madison: ASA, SSSA.
- McBride, M. B. (1998). Growing food crops on sludge-amended soils: Problems with the U.S. environmental protection agency method of estimating toxic metal transfer. *Environmental Toxicology and Chemistry*, 17(11), 2274–2281.
- Müller, H., & Pröhl, G. (1993). ECOSYS-87: A dynamic model for assessing radiological consequences of nuclear accidents. *Health Physics*, 64(3), 232–252.
- Ng, Y. C. (1982). A review of transfer factors for assessing the dose from radionuclides in agricultural products. *Nuclear Safety*, 23(1), 57–71.
- Oswald, S., Kinzelbach, W., Greiner, A., & Brix, G. (1997). Observation of flow and transport processes in artificial porous media via magnetic resonance imaging in three dimensions. *Geoderma*, 80, 417–429.
- Phillips, R. E., Quisenberry, V. L., & Zeleznik, J. M. (1995). Water and solute movement in an undisturbed, macroporous column: Extraction pressure effect. *Soil Science Society of America Journal*, 59, 707–712.
- Pierret, A., Moran, C. J., & Pankhurst, C. E. (1999). Differentiation of soil properties related to the spatial association of wheat roots and soil macropores. *Plant and Soil*, 211, 51–58.
- Renaud, P., Stapel, R., Maubert, H., Bleher, M., & Wirth, E. (1999). Comparative study of the PARK and ASTRAL post-accident decision support software. *Health Physics*, 76(5), 502–509.
- Roth, K., Schulin, R., Flüher, H., & Attinger, W. (1990). Calibration of time domain reflectometry for water content measurement using a composite dielectric approach. *Water Resources Research*, 26(10), 2267–2273.
- Schaap, M. G., Leij, F. J., & van Genuchten, M. T. (1998). Neural network analysis for hierarchical prediction of soil hydraulic properties. *Soil Science Society of America Journal*, 62, 847–855.
- Sharma, A. K., Srivastava, P. C., & Johri, B. N. (1994). Contribution of VA mycorrhiza to zinc uptake in plants. In J. A. Manthley, D. E. Crowley, & D. G. Luster (Eds.), *Biochemistry of metal micronutrients in the rhizosphere* (pp. 111–123). Boca Raton: Lewis Publishers.
- Shuman, L. M., & McCracken, D. V. (1999). Tillage, lime and poultry litter effects on plant concentrations of zinc, manganese, and copper. *Journal of Plant Nutrition*, 22(3), 609–620.
- Sinaj, S., Mächler, F., & Frossard, E. (1999). Assessment of isotopically exchangeable Zinc in polluted and nonpolluted soils. *Soil Science Society of America Journal*, 63, 1618–1625.
- Smolders, E., Brans, K., Földi, A., & Merckx, R. (1999). Cadmium fixation in soils measured by isotope dilution. *Soil Science Society of America Journal*, 63(1), 78–85.
- Smolders, E., van den Brade, K., & Merckx, R. (1997). Concentrations of ^{137}Cs and K in soil solution predict the plant availability of ^{137}Cs in soils. *Environmental Science and Technology*, 31, 3432–3438.
- Stewart, J. B. (1997). *The spatial distribution of plant roots and their interaction with soil structure*. Ph.D. dissertation, University of Sydney.
- Tardieu, F. (1991). Spatial arrangement of maize roots in the field. In: B. L. McMichael (Ed.), *Plant roots and their environment*, Vol. 24 (pp. 506–514). Developments in agricultural and managed-forest ecology, Amsterdam: Elsevier.

- Wadey, P., Shaw, G., Bell, J. N., & Minski, M. J. (1994). Radionuclide transport above a near-surface water table: II. Vertical distribution of gamma activities within soil profiles in relation to wheat rooting density and soil-to-plant transfers. *Journal of Environmental Quality*, 23, 1330–1337.
- Whiteley, G. M., & Dexter, A. R. (1983). Behaviour of roots in cracks between soil peds. *Plant and Soil*, 74, 153–162.